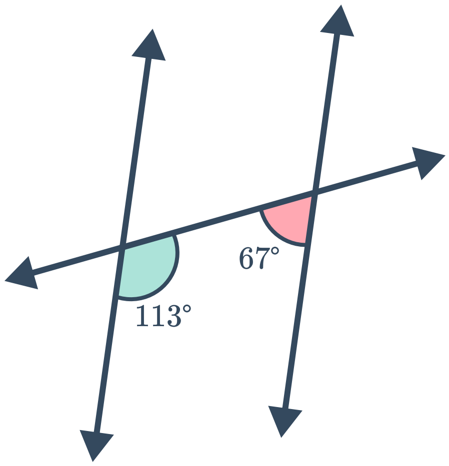
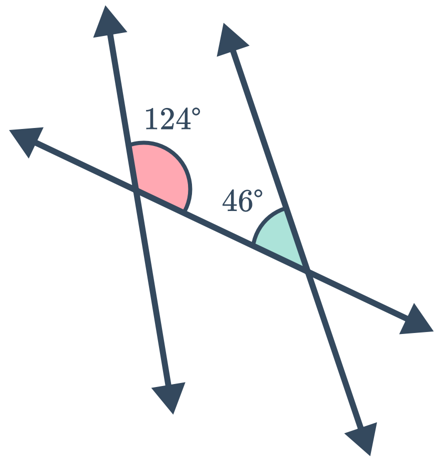
## Proving lines parallel

**1** For the following, state whether the diagram contains parallel lines. Justify your choice with a theorem or postulate.

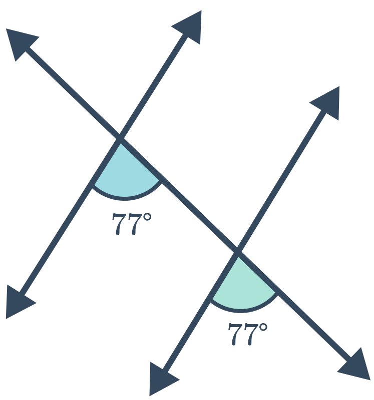
**a**



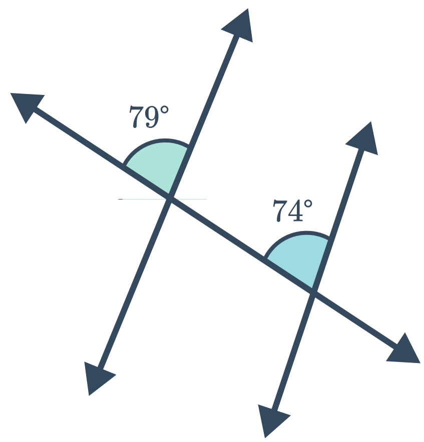

**b**



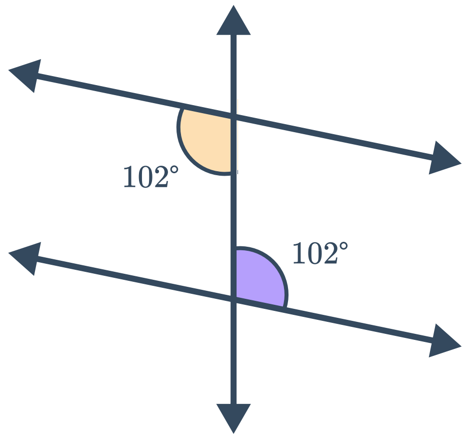

**c**

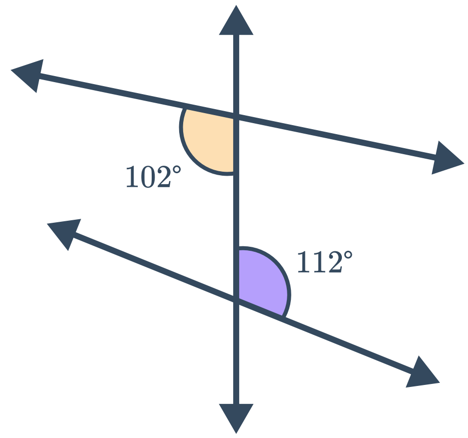



**d**

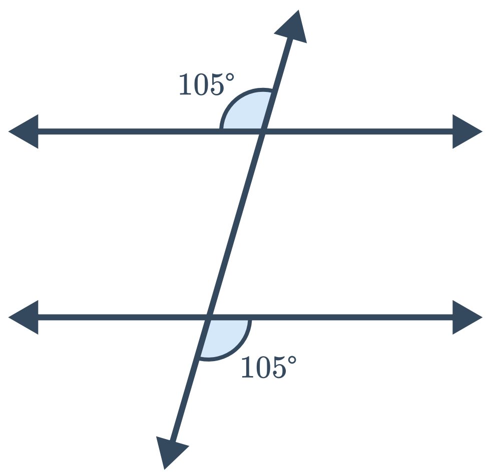


e



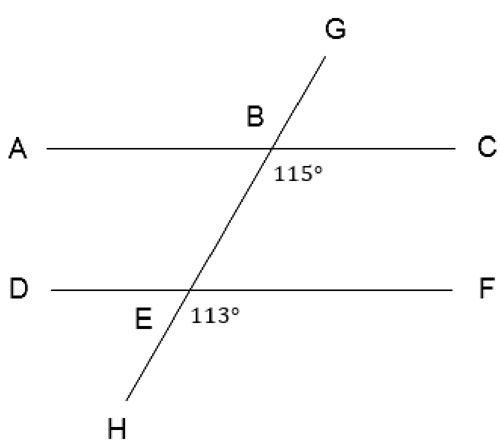




g

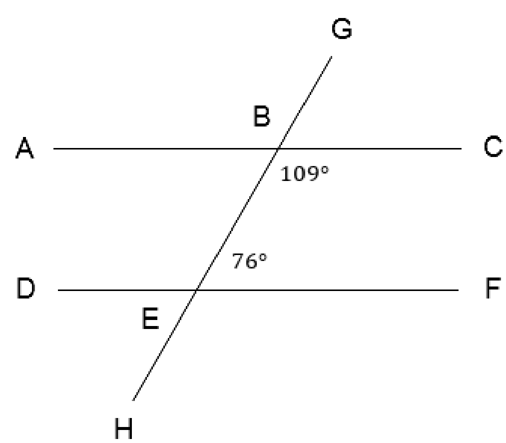



2 For the following diagrams, state whether $\overline{AC}$ and $\overline{DF}$ are parallel. Explain your reasoning.

a

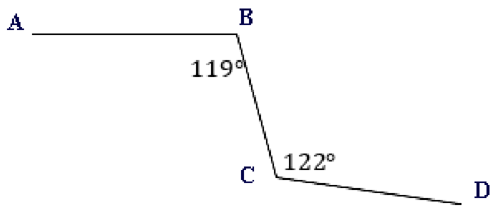



b

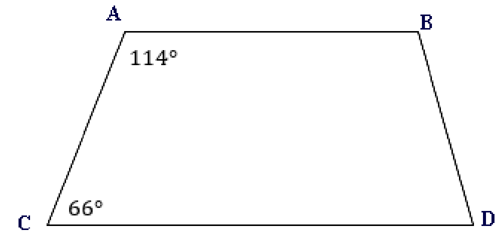


3 For the following diagrams, state whether [redacted]d $\overline{CD}$ are parallel. Explain your reasoning.

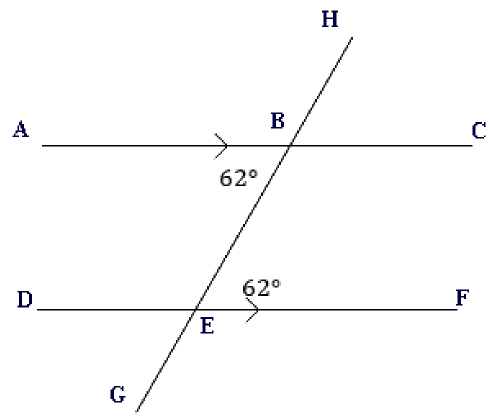
a




b




4 Eileen was given the following diagram, in which $m\angle ABE = m\angle BEF = 62^\circ$. She immediately marked the parallel symbols on the diagram to show that $\overline{AB} \parallel \overline{DE}$ . State whether the following reasons justify Eileen's markings.




a The two congruent angles form a pair of corresponding angles. Since the corresponding angles are congruent, $\overline{AB}$ must be parallel to $\overline{DE}$.

b The diagram is drawn to scale and so the lines look to be parallel.

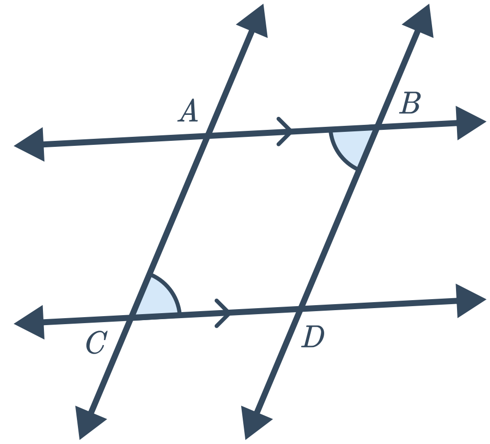
c The two congruent angles form a pair of alternate interior angles. Since the alternate interior angles are congruent, $\overline{AB}$ must be parallel to $\overline{DE}$.

d The two congruent angles form a pair of consecutive interior angles. Since the consecutive interior angles are congruent, $\overline{AB}$ must be parallel to $\overline{DE}$.

5 Suppose that two lines, $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$, are intersected by a transversal. Which of the following reasons alone could be used to show that the lines are parallel?

- Consecutive interior angles converse
- Alternate exterior angles theorem
- Linear pair theorem
- Congruent supplements theorem

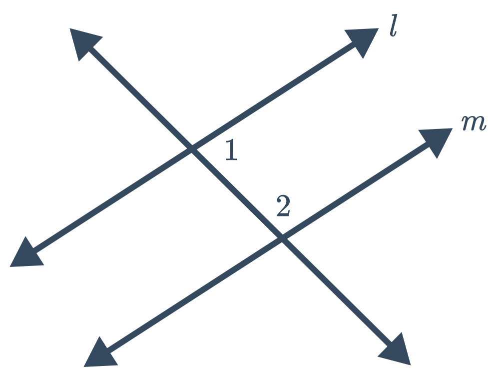
6 The following diagram shows that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ and $\angle ABD \cong \angle DCA$. Prove that $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$.



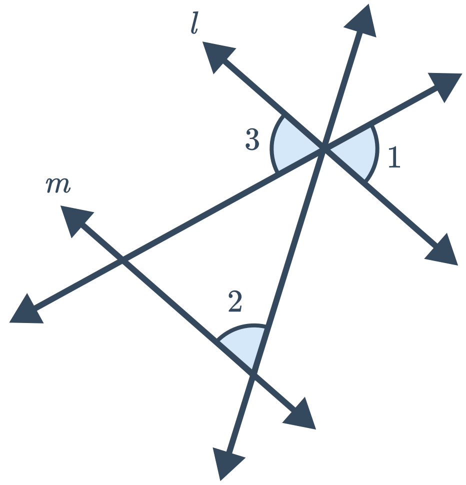

## Additional questions

7 Review the statement. Describe the error.

- If $\angle 1 \cong \angle 2$, then by the converse of the consecutive interior angles theorem, $l \parallel m$.

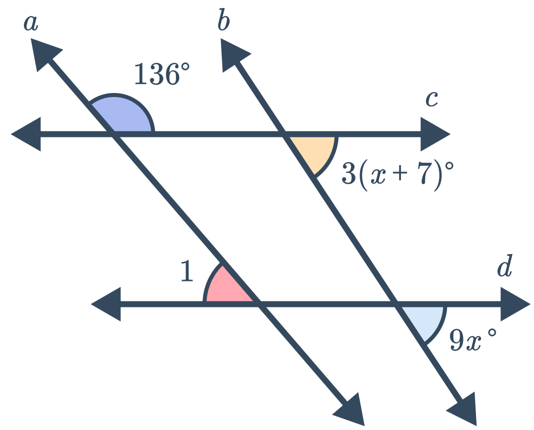


**8** Review the following proof. Describe the error.
$\angle 1 \cong \angle 2$ is given. By the vertical angles theorem, $\angle 1 \cong \angle 3$. So by the transitive property of congruence, $\angle 2 \cong \angle 3$. By the converse of the corresponding angles theorem, $l \parallel m$



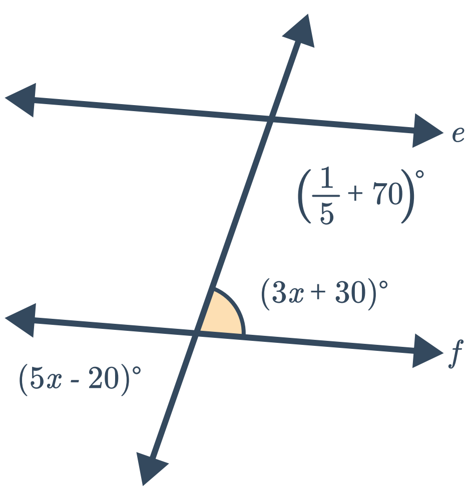

**9** Consider the given diagram:

- **a** Find the value of $x$ such that $c \parallel d$.
- **b** Find $m\angle 1$ such that $c \parallel d$. Justify your reasoning.

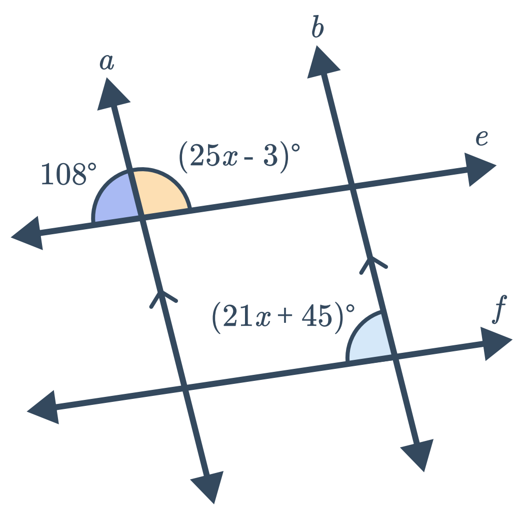



**10** Determine the value of $x$ that makes $e \parallel f$.

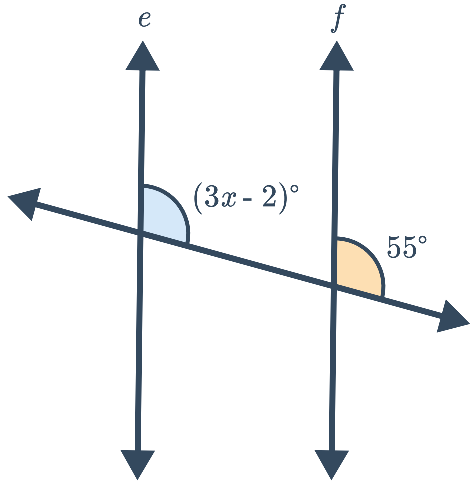
**a**

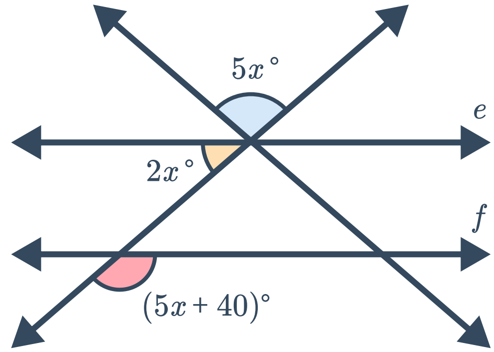



**b**

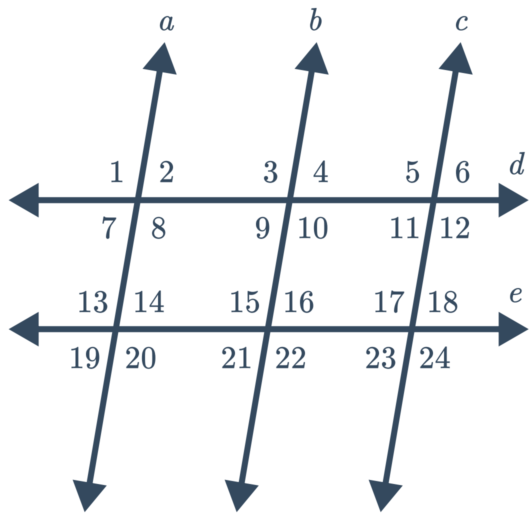


**C**



**11** Consider the following diagram for the below statements. Determine if the information given is enough to justify the conclusion. Explain your reasoning.




**a** Given:

- $\angle 1$ and $\angle 14$ are supplementary angles
- $\angle 22 \cong \angle 1$

Conclusion: $a \parallel b$

**b** Given:

- $\angle 7$ and $\angle 10$ are supplementary

Conclusion: $b \parallel c$

**c** Given:

- $b \parallel c$
- $\angle 5$ and $\angle 19$ are supplementary

Conclusion: $a \parallel b$

**d** Given:

- $\angle 1$ and $\angle 11$ are supplementary

Conclusion: $a \parallel c$

**e** Given:

- $\angle 18$ and $\angle 20$ are supplementary
- $a \parallel b$

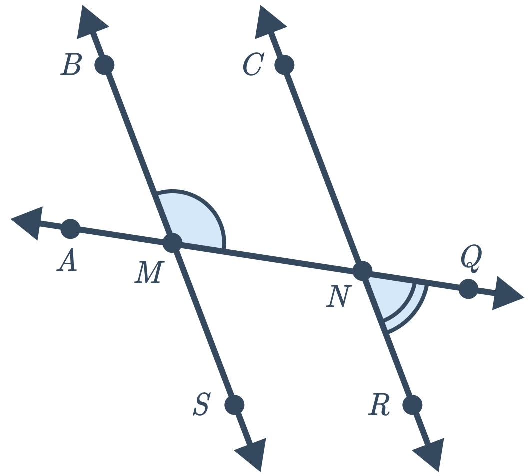
Conclusion: $b \parallel c$

**f** Given:

- $\angle 1 \cong \angle 13$
- $\angle 3 \cong \angle 24$

Conclusion: $a \parallel b$

**12** Consider the following diagram:




**a** Identify the angles that are supplementary to $\angle BMN$.

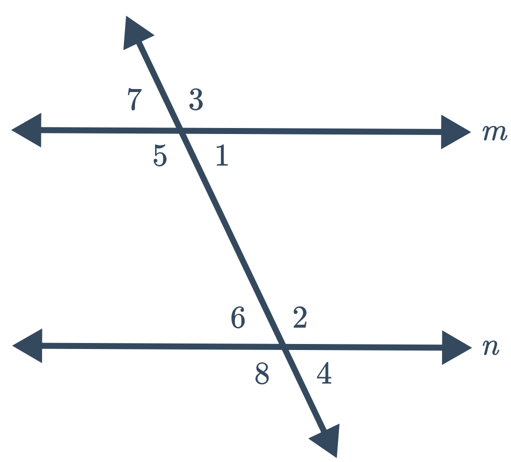
**b** If $\angle BMN$ and $\angle QNR$ are supplementary, prove $\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$.

**13** Given:

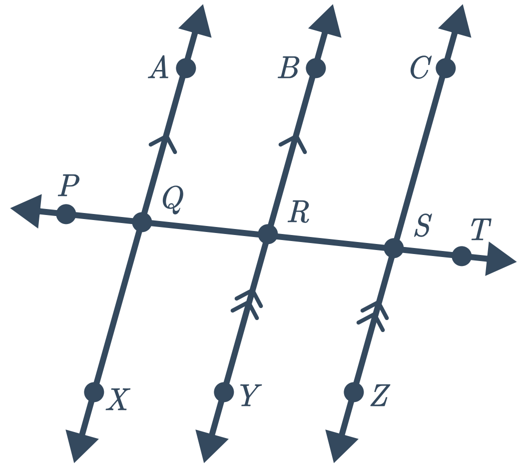
- $\angle A$ and $\angle D$ are supplementary
- $\angle B \cong \angle D$

Prove: $\overline{AD} \parallel \overline{BC}$

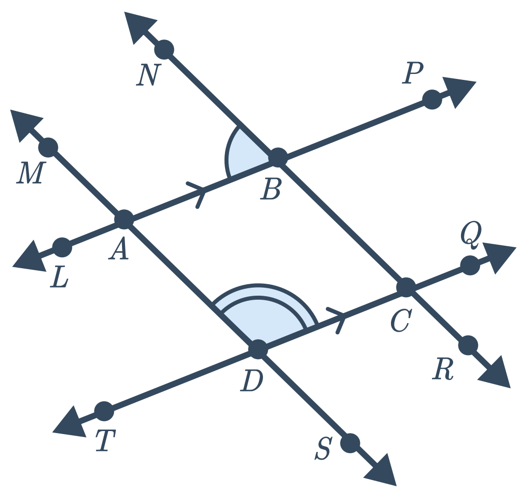

**14** Prove the converse of the consecutive interior angles theorem:
Given: $\angle 1$ is supplementary to $\angle 2$
Prove: $m \parallel n$



**15** Given: $\overleftrightarrow{AX} \parallel \overleftrightarrow{BY}$ and $\overleftrightarrow{BY} \parallel \overleftrightarrow{CZ}$
Prove: $\angle CST \cong \angle XQP$




**16** Given: $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ and $\angle NBA$ and $\angle ADC$ are supplementary
Prove: $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$




**17** Given: $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ and $\angle MAB \cong \angle BCQ$
Prove: $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$